

TS6 — Lean 4 Formalization Layer for the TS Programme on the Smoothed Prime Variance

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Abstract

This short note is the dedicated formalization layer of the TS programme (TS1–TS5) on the empirically-centered variance of smoothed prime sums in arithmetic progressions. TS6 formalizes, in Lean 4 over Mathlib4, the four purely algebraic identities that constitute the exact kernel of the programme: the abstract Parseval identity on a finite abelian group, its specialization to $(\mathbb{Z}/q\mathbb{Z})^\times$, the centering-shift identity, and the Rankin–Selberg decomposition. The effective layer (TS3, TS4) is scaffolded conditionally on two named hypothesis structures (large sieve, effective Chebyshev) that Mathlib4 does not yet provide; no `axiom` is used anywhere. TS6 is delivered here in YELLOW status: every Lean statement is precisely typed and its proof plan is documented, but the proof-term bodies remain `sorry` because the sandbox in which this note was produced does not permit installation of a Lean toolchain. Closing the exact-kernel `sorry`s to reach GREEN is a routine exercise of a few hours in a live Mathlib4 environment. This note states the scope, the architecture, the target theorems, the current limits, and the reproducibility instructions.

1 Scope

The TS programme (TS1–TS5) organizes results about the quantity

$$\mathcal{A}_W^*(x, q) := \sum_{\substack{a=1 \\ (a, q)=1}}^q |\theta_W(x; q, a) - \bar{\theta}_W(x, q)|^2,$$

the empirically-centered variance of the smoothed prime sum $\theta_W(x; q, a) = \sum_{p \equiv a (q)} (\log p) W(p/x)$ across invertible residue classes modulo q . The programme’s three layers are:

Exact. $\mathcal{A}_W^*(x, q) = \mathcal{C}_W(x, q)$ for every integer $q \geq 2$, and the centering-shift identity relating $\mathcal{A}_W^{\text{PNT}}$ to $\mathcal{C}_W$.

Effective. $\sum_{q \leq Q, q \text{ prime}} \mathcal{A}_W^*(x, q) \leq 2\|W\|_\infty^2 (Q^2 + 2x) x \log(2x)$ via the large sieve and Chebyshev.

Open. Three asymptotic questions, precisely scoped.

TS6 targets *only* the exact layer of that classification. The aim is to obtain a Lean 4 formal certificate that the four exact identities hold, as a final piece of documentation for the programme.

What this note does not do. TS6 is deliberately narrow. It does not formalize the empirical observations of TS1 §5 or TS5 §3 (regression fits, the factor-5 diagonal-overshoot observation). It does not attempt to close the effective layer: the large sieve inequality and the effective Chebyshev bound $\vartheta(y) \leq 2y$ are not presently in Mathlib4 in the required form, and TS6 refuses to discharge them by `axiom`. It does not formalize any of the three open problems of TS5 §5. These exclusions are stated here once and enforced throughout.

2 Exact kernel in Lean 4

The central targets are named TS6-A through TS6-D.

TS6-A — Abstract Parseval on a finite abelian group

Let G be a finite abelian group and $f : G \rightarrow \mathbb{C}$. Define the empirical mean $\mu(f) = |G|^{-1} \sum_g f(g)$, the variance $\text{Var}(f) = \sum_g |f(g) - \mu(f)|^2$, the Fourier coefficient $\hat{f}(\chi) = \sum_g f(g) \overline{\chi(g)}$, and the non-principal character-side mean square $\widehat{C}(f) = |G|^{-1} \sum_{\chi \neq 1} |\hat{f}(\chi)|^2$.

Theorem 2.1 (TS6-A, abstract Parseval identity). *For every finite abelian group G and every $f : G \rightarrow \mathbb{C}$,*

$$\text{Var}(f) = \widehat{C}(f).$$

The proof is the standard one-line expansion using Schur orthogonality on G . The Lean target is the file `TS6/Exact/FiniteCharacterVariance.lean`, which imports `Mathlib.NumberTheory.MulChar.Basic` and uses the lemma `MulChar.sum_apply_eq_zero_iff` for non-trivial characters.

TS6-B — Modular specialization

Specializing Theorem 2.1 to $G = (\mathbb{Z}/q\mathbb{Z})^\times$ and $f(r) = \theta_W(x; q, r)$ (viewed as a function of a class) yields:

Theorem 2.2 (TS6-B, exact identity for general q , TS2 Thm 2.1). *For every integer $q \geq 2$, every $x > 0$, and every bounded weight W with compact support in $(0, \infty)$,*

$$\mathcal{A}_W^*(x, q) = \mathcal{C}_W(x, q).$$

The Lean target is `TS6/Exact/DirichletVariance.lean`. The proof is a direct application of TS6-A through the bijection $\text{MulChar}(\mathbb{Z}/q)^\times \mathbb{C} \longleftrightarrow \text{DirichletCharacter } \mathbb{C}^q$.

TS6-C — Centering-shift identity

The centering-shift identity compares the empirical-mean centering to the PNT-predicted centering $x\widehat{W}(1)/\varphi(q)$.

Theorem 2.3 (TS6-C, centering-shift identity, TS2 Prop 2.2). *Under the hypotheses of Theorem 2.2,*

$$\mathcal{A}_W^{\text{PNT}}(x, q) - \mathcal{C}_W(x, q) = \frac{|\Theta(x, \chi_0) - x\widehat{W}(1)|^2}{\varphi(q)}.$$

The proof is pure shift-of-origin for the variance of a finite sequence: $\sum_r |t_r - \mu|^2 = \sum_r |t_r - \bar{t}|^2 + |G| |\bar{t} - \mu|^2$, applied at $\mu = x\widehat{W}(1)/\varphi(q)$ and combined with Theorem 2.2. The Lean target is `TS6/Exact/CenteringShift.lean`.

TS6-D — Rankin–Selberg decomposition

Writing $a_p = (\log p) W(p/x)$ and $T_0(x, q) = \sum_{(p,q)=1} a_p$, define the diagonal $\mathcal{D}_W(x, q) = \sum_{(p,q)=1} |a_p|^2$ and the off-diagonal

$$\mathcal{S}_{Wq}(x) = \sum_{\substack{p_1 \neq p_2 \\ p_1 \equiv p_2 \pmod{q} \\ (p_1 p_2, q)=1}} a_{p_1} \overline{a_{p_2}}.$$

Theorem 2.4 (TS6-D, Rankin–Selberg decomposition, TS5 Prop 3.7). *For every integer $q \geq 2$,*

$$\mathcal{A}_W^*(x, q) = \mathcal{D}_W(x, q) - \frac{|T_0(x, q)|^2}{\varphi(q)} + \mathcal{S}_{Wq}(x).$$

This identity is the algebraic output of expanding $\mathcal{C}_W$ via the character-sum form of Theorem 2.2 and splitting the resulting prime-pair double sum into $p_1 = p_2$ (diagonal) vs. $p_1 \neq p_2$ (off-diagonal) contributions, subtracting the principal-character square. The Lean target is `TS6/Structure/RankinSelbergDecomp.lean`. No analytic input is used: the identity is structural.

3 Modular specialization in code

The core of TS6-B in Lean has the following shape (simplified):

```
variable (q : N) [NeZero q] (a : N -> C) (S : Finset N)

noncomputable def theta (r : (ZMod q)^x) : C :=
  sum n in S, if (n : ZMod q) = (r : ZMod q) then a n else 0

noncomputable def theta_bar : C :=
  (sum r : (ZMod q)^x, theta a q S r) / (totient q : C)

noncomputable def AWstar : R :=
  sum r : (ZMod q)^x, norm (theta a q S r - theta_bar a q S)^2

noncomputable def CW : R :=
  (1 / (totient q : R)) *
    sum chi in (univ : Finset (DirichletCharacter C q)).filter (. ne 1),
      norm (sum n in S, a n * chi n)^2

theorem TS6_B_exact_identity
  (hq : 2 <= q) : AWstar a q S = CW a q S := by
  -- specialize parseval_identity to G = (ZMod q)^x
  -- carry coefficients through the MulChar ~ DirichletCharacter bijection
  sorry
```

The single `sorry` is the specialization step: every piece is in place to push Theorem 2.1 through the isomorphism and rename.

4 Centering-shift in code

TS6-C is packaged as a separate module because it depends on TS6-B but is otherwise pure linear algebra over $\mathbb{C}$. The core shift-of-origin lemma is this:

```
-- For any finite-indexed complex sequence (t_i) and any target mu in C,
--   sum i, ||t i - mu||^2
--   = sum i, ||t i - (sum j, t j) / I||^2 + |I| * ||mu - (avg)||^2.
lemma variance_shift_of_origin
  {I : Type*} [Fintype I] (t : I -> C) (mu : C) :
  sum i, norm (t i - mu)^2 =
    sum i, norm (t i - (sum j, t j) / (Fintype.card I : C))^2
    + (Fintype.card I : R) *
      norm (mu - (sum j, t j) / (Fintype.card I : C))^2 := by
  ring_nf -- after expanding norms via star multiplication
  sorry
```

TS6-C follows by instantiating this at $I = (\mathbb{Z}/q)^\times$, $t_r = \theta_W(x; q, r)$, and $\mu = x\widehat{W}(1)/\varphi(q)$, then using TS6-B to replace the first sum on the right by $\mathcal{C}_W(x, q)$ and computing the second term as $|\Theta(x, \chi_0) - x\widehat{W}(1)|^2/\varphi(q)$.

5 Current limits of formalization

TS6 does *not* claim to have closed any Lean proof in the current session: **the Lean 4 toolchain is not available in the environment where this paper is produced**. Every statement in the Lean files `TS6/Exact/FiniteCharacterVariance.lean`, `TS6/Exact/DirichletVariance.lean`, `TS6/Exact/CenteringShift.lean`, and `TS6/Structure/RankinSelbergDecomp.lean` is typed and its proof sketch is documented in a comment block, but the proof-term bodies are `sorry`.

No `axiom` is used anywhere. The statements are proof-plan-complete; the tactic scripts to close them are routine in a live Mathlib4 environment and are marked as future work.

The effective-layer statements in `TS6/Effective/AveragedBounds.lean` are conditional on two typed hypothesis structures `LargeSieveInequality` and `ChebyshevThetaBound` — not `axioms` — and will remain so until those inputs land in Mathlib4.

Summary table.

Target	TS1–5 source	Lean file	Current status
TS6-A	kernel for TS1, TS2	<code>Exact/FiniteCharacterVariance</code>	YELLOW
TS6-B	TS2 Thm 2.1	<code>Exact/DirichletVariance</code>	YELLOW
TS6-C	TS2 Prop 2.2	<code>Exact/CenteringShift</code>	YELLOW
TS6-D	TS5 Prop 3.7	<code>Structure/RankinSelbergDecomp</code>	YELLOW
TS6-E	TS3 Thm 1.1	<code>Effective/AveragedBounds</code>	YELLOW (by design)
TS6-F	TS4 Thm 1.1	<code>Effective/AveragedBounds</code>	YELLOW (by design)
TS6-G	TS4 Cor 1.2	<code>Effective/AveragedBounds</code>	YELLOW (by design)

YELLOW means: statement typed, proof plan written, no `axiom`, `sorry` explicit. The target for a subsequent session with a Lean toolchain available is GREEN on TS6-A through TS6-D. TS6-E through TS6-G stay YELLOW until Mathlib supplies the two missing analytic inputs; they will not become GREEN in any near-term TS6 iteration.

6 Reproducibility / build instructions

To verify or extend this formalization layer in a live Lean environment:

1. Install `elan` and Lean 4 per the `leanprover-community` install guide.
2. Create a Mathlib-based Lean project with `lake new TS6 math`.
3. Copy the `lean/TS6/` directory of this bundle into `TS6/TS6/` inside the new project.
4. Add `import TS6.Exact.FiniteCharacterVariance` (and the three other main files) to the project’s root file.
5. Run `lake build`. Each `sorry` will be flagged; closing them is the formalization work.

The hash manifest `TS6_FROZEN.txt` in the bundle records the SHA-256 hash of each file in TS6 as released.

Labels and honesty

All claims in this note are labelled by their logical type, consistent with the TS programme’s discipline:

- Theorems TS6-A, TS6-B, TS6-C, TS6-D are **exact identities**. Their proofs in the paper setting are part of TS1–TS5 and are independently verified there at machine precision on 672 prime- q and 126 composite- q triples (see TS1 §5 and TS2 §4 of the merged bundle). Their Lean formalization status is YELLOW as described.
- TS6-E, TS6-F, TS6-G are **effective theorems** whose paper-side proofs are in TS3 and TS4, and whose Lean status is YELLOW by design pending Mathlib upgrades.
- No **empirical observation** or **open problem** is formalized or referenced by TS6.

TS6 does not close the TS programme: the asymptotic law for $\mathcal{A}_W^*$ remains the programme’s principal open question. TS6 locks the exact-kernel identities at the level of typed Lean statements with documented proof plans, producing a certified index of what the TS programme’s formal backbone looks like.